

11. Which of the following soils is most suited for tea plantation?

- (a) Acidic
- (b) Alkaline
- (c) Alluvial
- (d) Regur

U.P.P.C.S. (Pre) 2002

Ans. (a)

Tea is grown in a variety of soils. The best, however, is soil slightly acidic and without calcium as tea is highly intolerant to stagnant water. Although tea requires heavy rainfall for its growth, stagnant water is injurious to its roots. It is, therefore, grown on hill slopes where water drains away easily and waterlogging does not take place. However, it grows equally well in the valley if the drainage system is good.

vi. Soil Erosion and Conservation

***Soil Erosion** - The destruction of the soil cover is described as soil erosion. Soil erosion refers to the **wearing away and removal of the topsoil** by natural physical forces such as water, wind and air. ***Topsoil** is the top layer of soil and is the most fertile because it contains the most organic, nutrient-rich substances.

Causes of Soil Erosion-

- (1) Deforestation
- (2) Disturbance in the natural drainage pattern, improper design of waterways.
- (3) Grazing, Water, Wind, Air
- (4) Poor vegetation and cultivation

*The most affected regions by soil erosion in India are **Badlands of the Chambal and Yamuna rivers**, Western Himalaya (Shivalik region), Chota Nagpur plateau, Malwa plateau, Black Soil region of Maharashtra, arid regions of Haryana, Rajasthan and Gujarat. Ravines of Chambal valley are formed by **Gully erosion**.

***Splash Erosion:** The erosion due to the impact of falling raindrops on soil-surface leading to the destruction of the crumb structure is known as the raindrop or splash erosion. This is the **first stage of erosion**.

***Sheet Erosion:** Sometimes water flows as a sheet over large areas down a slope. In such cases, the topsoil is washed away. This is known as sheet erosion. It is the **second stage of soil erosion**.

***Rill Erosion :-** This is the **third stage of sheet erosion**. If erosion continues unchecked for a sufficient time then rills or small finger-shaped grooves which are a few centimetres in

depth may develop on the landscape. Over a period of time, the fine rills increase in number and also become deeper and wider, resembling the twigs, branches and trunk of a tree. This is called rill erosion.

***Gully Erosion :-** This is the **fourth stage of erosion**. With further erosion of the soil, the rills may become deeper and enlarge the (region between the Tapi and Sabarmati valleys, and are ultimately turning into gullies. Gully erosion is also responsible for the formation of ravines.

***Stream Channel Erosion :-** Water flowing through a channel corrodes its banks and base until a stable base is formed. This form of erosion is known as stream channel erosion.

***Soil Conservation :-** Prevention of soil erosion is also called conservation of soil. The soil erosion can be prevented by promoting afforestation, **Crop Rotation, Terrace Farming, Building Dams, Shelter Belts, Ebankments**, etc.

***Crop rotation** refers to the practice of growing different types of crops over the same area in different seasons. It enriches soil fertility and helps in pest control. Plants keep soil bound in locks of roots, and thus, prevent soil erosion.

***Reduced or zero tillage** of the land in farming is a way of growing crops or pasture year after year without disturbing the soil through tillage. It helps in soil moisture conservation.

1. In which one of the following areas in India, is the problem of soil erosion is critical?

- (a) Malwa Plateau
- (b) Foothill areas of Shivalik ranges
- (c) Dandkaranya
- (d) Meghalaya Plateau

U.P.P.C.S. (Pre) 2004

Ans. (b)

The maximum soil erosion affected areas in India are given below :

- 1. Barren land of Chambal and Yamuna rivers
- 2. Foothill areas of Western Himalayan Ranges (including foothill areas of the Shiwalik Ranges)
- 3. Chotanagpur Plateau
- 4. The area between Tapi and Sabarmati Valley (Malwa Plateau, etc.)
- 5. Black soil area of Maharashtra
- 6. Arid regions of Haryana, Rajasthan & Gujarat.

2. Which one of the following areas is most affected by soil erosion in India?

- (a) Malwa Plateau
- (b) Tarai area of Uttar Pradesh
- (c) Coastal area of Andhra
- (d) Chambal valley

M.P.P.C.S. (Pre) 2006

U.P. Lower Sub. (Pre) 2004

Ans. (d)

See the explanation of the above question.

3. Which among the following districts of Madhya Pradesh has a problem of soil erosion -

- (a) Jabalpur
- (b) Bhopal
- (c) Morena
- (d) Khandwa

M.P.P.C.S. (Pre) 2008

Ans. (c)

In Madhya Pradesh, the problem of soil erosion is mostly found in Chambal valley where channel formation is very common and known as Chambal ravines. Soil contents are characterized by soft and alluvial soil, which help in the formation of these channels. Mainly Morena, Bhind, Gwalior and Shivpuri have become barren and infertile due to gully erosion by various rivers.

4. Which one of the following types of erosion is responsible for the formation of Chambal Ravines?

- (a) Splash
- (b) Sheet
- (c) Rill
- (d) Gully

I.A.S. (Pre) 1994

Ans. (d)

After viewing the map of soil erosion in India, it is clear that Gully Erosion is responsible for the formation of Chambal Ravines.

5. Identify the correct order of the process of soil-erosion from the following:

- (a) Splash erosion, Sheet erosion, Rill erosion, Gully erosion
- (b) Sheet erosion, Splash erosion, Gully erosion, Rill erosion,
- (c) Rill erosion, Gully erosion, Sheet erosion, Splash erosion
- (d) Gully erosion, Rill erosion, Sheet erosion, Splash erosion

I.A.S. (Pre) 2001

Ans. (a)

The correct order of the process of soil erosion is given below:

i. **Raindrop or Splash erosion:** Erosion preceded by the destruction of the crumb structure due to the impact of falling raindrops on the surface of the soil is termed as splash erosion.

ii. **Sheet erosion:** It is the fairly uniform removal of soil in thin layers from the land surface, often scarcely perceptible, especially when caused by wind. Areas where loose, shallow topsoil overlies compact soil are most susceptible to sheet erosion.

iii. **Rill erosion:** A form of water erosion in which numerous very small and more or less straight channels are produced; the channels get obliterated by ordinary use. It can be removed by normal tillage operations.

iv. **Gully erosion:** A form of water erosion in which gullies are produced by a combination of unattended rills.

v. **Stream channel erosion:** Stream banks are eroded by water either flowing over the sides of a stream or scouring at the base. It is aggravated by the removal of vegetation, overgrazing, or cultivation near the stream banks.

6. Farmland tends to erode most rapidly when planted with:

- (a) Sorghum
- (b) Potato
- (c) Wheat
- (d) Clover

I.A.S. (Pre) 1994

Ans. (a)

According to an analysis by the World Wildlife Fund, farm land tends to erode most rapidly when planted with sorghum while clover causes minimum soil erosion among the given options.

7. Which of the following is not a correct method to attain soil fertility?

- (a) Crop rotation
- (b) Mixed cultivation
- (c) Seed modification
- (d) Multiple cropping

M.P. P.C.S. (Pre) 1991

Ans. (c)

The seed modification method would affect crop production but not affect soil fertility. Soil fertility can be improved by crop rotation, mixed cultivation and multiple cropping methods.

8. The rotation of crops is essential for –

- (a) Increasing protein content in plants
- (b) Getting different crops
- (c) Increasing soil fertility
- (d) Maintaining soil moisture

Uttarakhand P.C.S. (Pre) 2010

Ans. (c)